

# Modulus Functions & the Factor/Remainder Theorem



## Evaluating and solving modulus equations

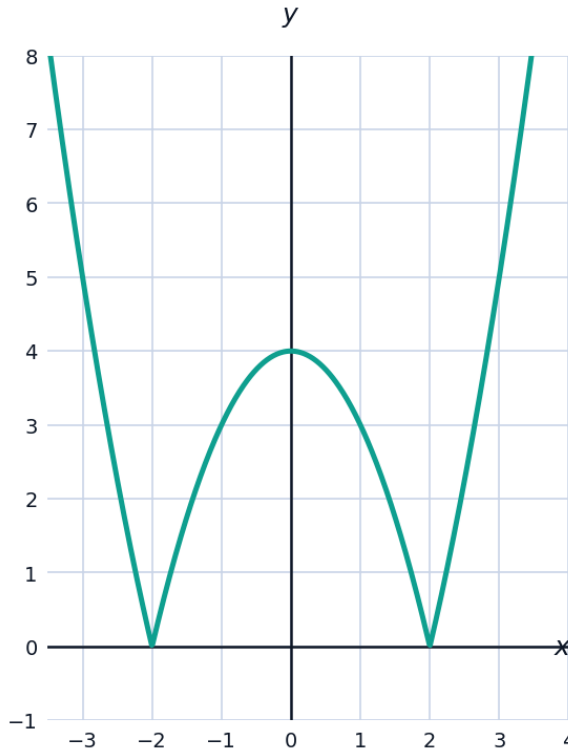
- 1 Solve  $|2x - 3| = 7$ .
  - 2 Solve  $|x - 4| = |2x + 1|$ .
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## Modulus inequalities

- 3 Solve the inequality  $|3x + 2| < 8$ , giving your answer as an interval.
  - 4 Solve  $|x - 5| \geq 3$ , giving your answer as a union of intervals.
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## Visualizing the graph of $y = |f(x)|$

- 5 The graph shows  $y = |x^2 - 4|$ .



- a Explain how this graph relates to the graph of  $y = x^2 - 4$ , and identify which portion has been reflected.
- b State the  $x$ -values where the graph touches the  $x$ -axis, and explain why the curve has sharp corners there rather than smooth turning points.

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### The factor theorem

- 6 Use the factor theorem to show that  $(x - 2)$  is a factor of  $f(x) = x^3 - 3x^2 - 4x + 12$ , then factor  $f(x)$  completely.
- 7 Find the value of  $k$  such that  $(x + 3)$  is a factor of  $f(x) = x^3 + kx^2 - 5x - 6$ .

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### The remainder theorem

- 8 Find the remainder when  $f(x) = 2x^3 - 5x^2 + 3x - 7$  is divided by  $(x - 2)$ .
- 9 When  $f(x) = x^3 + ax^2 + bx - 4$  is divided by  $(x - 1)$  the remainder is 2, and when divided by  $(x + 2)$  the remainder is  $-22$ . Find  $a$  and  $b$ .

## Solving equations involving modulus functions graphically

- 10 Solve  $|x - 3| = 2x + 1$ , checking that each solution satisfies the original equation (since squaring or case-splitting can introduce extraneous solutions).
- 11 Solve the inequality  $|2x + 3| \leq |x - 1|$  by squaring both sides (valid since both sides are non-negative).
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## Finding all zeros of a higher-degree polynomial

- 12 Given that  $x = 1$  and  $x = -1$  are both zeros of  $f(x) = x^4 - 5x^3 + 5x^2 + 5x - 6$ , find the remaining two zeros.
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## Using the remainder theorem to test multiple divisors at once

- 13 A polynomial  $f(x)$  leaves remainder 5 when divided by  $(x - 2)$  and remainder  $-3$  when divided by  $(x + 1)$ . Find the remainder when  $f(x)$  is divided by  $(x - 2)(x + 1)$ , given that the remainder must be of the form  $ax + b$ .
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## Solving polynomial equations with the factor theorem

- 14 Solve  $2x^3 - x^2 - 13x - 6 = 0$  completely, given that  $x = -2$  is one root.
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## Graphing modulus functions with restrictions

- 15 Sketch, in words, the key features of  $y = |2x - 4| + 1$ : state its minimum value, the  $x$ -value where it occurs, and its  $y$ -intercept.
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## Finding unknown coefficients using both theorems together

- 16 The polynomial  $f(x) = x^3 + ax^2 + bx + 6$  has  $(x - 1)$  as a factor and leaves a remainder of  $-12$  when divided by  $(x + 2)$ . Find  $a$  and  $b$ .
- 17 Show that  $f(x) = x^3 - 6x^2 + 11x - 6$  has three distinct real roots by testing  $x = 1, 2, 3$  using the factor theorem, and write  $f(x)$  in fully factored form.